

MP1994V2 Milestone Tracker

Mortensen–Pissarides (1994) Lean replication

4 August 2026

Current milestone: M10.1 – Finite-state Markov equilibrium, equations (32)–(34)

Current status: NOT STARTED

Next action: Implement the finite-state continuous-time representation

Last completed milestone: M9.3 – Job creation, unemployment dynamics, and impact asymmetry –
COMPLETE – GREEN

Source of truth. docs/milestone_tracker.md. Status vocabulary: NOT STARTED; IN PROGRESS; BLOCKED; BLOCKED – CUTOFF ORDERING; READY FOR REVIEW; PARTIAL – READY FOR REVIEW; COMPLETE – GREEN; COMPLETE – AMBER; DEFERRED / OUTSIDE CORE LEAN. M0 through M4 are complete and green. M5 is complete and amber: uniqueness is green and existence is conditional on an additional bracket. M5b is unstarted. M6 is complete and amber overall: its stock completion and uniqueness are green, while existence inherits M5’s amber qualification. M7 is complete and green after human mathematical/economic review. M8 is COMPLETE – AMBER after review on 2026-08-01: analytic infrastructure is GREEN and supplied differentiable local paths cause the AMBER qualification. This is not inherited from M5. That historical grade applies to the original supplied-path layer. M8b.1 and M8b.2 derive all four paths from a supplied regular reduced equilibrium, so the full Appendix package is COMPLETE – GREEN under explicit Appendix matching/IFT regularity and sign conditions. Primitive selection of the base equilibrium would separately inherit M5’s AMBER existence qualification.

Milestone plan

The tracker uses one readable block per milestone instead of a dense landscape table. Each block contains the same fields as the Markdown table.

M0. Foundations and primitive value equilibrium

Status: COMPLETE – GREEN

Paper: Sections 2–3; equations (1)–(7)

Economic target: Primitives, layered assumptions, probability/matching definitions, and primitive value equilibrium.

Expected declarations: Primitives; four assumption bundles; cdf; meeting rates; surplus; continuationTerm; ValueEquilibrium

Adequacy grade: GREEN **Dependencies:** None

Blocker: None

Branch/commit or tag; last review; next action: issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed M0 interface while beginning M1 separately.

M1. Surplus Bellman equation

Status: COMPLETE – GREEN

Paper: Section 3, journal p. 401; equation (8)

Economic target: Derive equation (8) from equations (3)–(7).

Expected declarations: Continuation decomposition; firm_share; surplus_flow_equation; surplus_bellman_of_probability; surplus_bellman

Adequacy grade: GREEN **Dependencies:** M0 complete

Blocker: None

Branch/commit or tag; last review; next action: issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed conditional representation theorem.

M2. Affine surplus and endogenous reservation cutoff**Status:** COMPLETE – GREEN**Paper:** Section 3; reservation discussion and equation (12)**Economic target:** Surplus difference theorem, strict monotonicity, unique endogenous reservation threshold, and equation (12).**Expected declarations:** Surplus difference, affinity, strict monotonicity, zero uniqueness, reservation theorem**Adequacy grade:** GREEN **Dependencies:** M1**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve M2 interfaces and design the measure-form continuation identity for M3.**M3. Continuation value, job destruction, and job creation****Status:** COMPLETE – GREEN**Paper:** Section 3; equations (9), (10), and (13)**Economic target:** Measure-form continuation identity and derivation of job destruction and job creation.**Expected declarations:** expectedExcess; tailOptionValue; equations (9), (10), and (13); JD/JC residual predicates; M3 capstone**Adequacy grade:** GREEN **Dependencies:** M2**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-30 human mathematical/economic review; preserve the independent JD/JC interfaces and neutral capstone module.**M4. Reduced-equilibrium reconstruction and equivalence****Status:** COMPLETE – GREEN**Paper:** Section 3; equations (10), (13); journal p. 402**Economic target:** Two-way representation between primitive values and the admissible reduced pair.**Expected declarations:** ReducedEquilibrium; generic tail identity; forward and reverse maps; round trips; nonempty equivalence**Adequacy grade:** GREEN **Dependencies:** M3**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-31 human mathematical/economic review; preserve representation interfaces and formulate corrected continuity, endpoint, and crossing assumptions for M5.**M5. Existence and uniqueness of static equilibrium****Status:** COMPLETE – AMBER**Paper:** Section 3; equations (10), (13), Figure 1, journal p. 402**Economic target:** Existence and uniqueness of the static reduced equilibrium.**Expected declarations:** Expected-excess regularity; JD/JC slopes; StaticExistenceAssumptions; conditional unique existence; value-side economic uniqueness**Adequacy grade:** Uniqueness: GREEN; existence: AMBER; overall: AMBER **Dependencies:** M4**Blocker:** Existence uses a lower positive crossing condition not derived from primitives**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-31 human review; preserve the conditional theorem and see the M5b design in docs/static_existence_foundation.md. This does not block M6.

M5b. Primitive foundation for static existence**Status:** NOT STARTED**Paper:** Foundational refinement to Section 3 static existence**Economic target:** Replace the assumed lower crossing with a derived result.**Expected declarations:** Proposed matching-boundary bundle; upper-job profitability; derived lower_crossing; optional Cobb–Douglas certification**Adequacy grade:** Target: existence GREEN under a strengthened general theorem or certified Cobb–Douglas specialization **Dependencies:** M5**Blocker:** The Inada condition on q alone is insufficient; JD-implied tightness must become positive through a separate profitability condition**Branch/commit or tag; last review; next action:** No branch/tag or review; add right-hand Inada and upper-job profitability, derive StaticExistenceAssumptions, and see docs/static_existence_foundation.md. M5b is not M6.**M6. Unemployment flow balance and full static steady state****Status:** COMPLETE – AMBER**Paper:** Section 3; equation (14), Figure 2, journal pp. 403–404**Economic target:** Derive unemployment, employment, vacancies, and the full static state.**Expected declarations:** Strict mass/CDF bridge; hazards and flows; steadyStateUnemployment; SteadyStateEquilibrium; equation (14); exact reduced/full equivalence**Adequacy grade:** Stock completion/equation (14): GREEN; uniqueness: GREEN; existence: AMBER inherited from M5; overall: AMBER **Dependencies:** M4–M5**Blocker:** No new M6 blocker. Full-state nonemptiness inherits M5’s unproved lower-crossing foundation**Branch/commit or tag; last review; next action:** issue2; M6 completion commit; 2026-07-31 human mathematical/economic review; preserve the split capstones and adequacy contract; M7 is NOT STARTED.**M7. Static order comparative statics****Status:** COMPLETE – GREEN**Paper:** Section 3, journal pp. 401–404; equations (10), (13), (14); Figures 1–2**Economic target:** Global order results for supplied equilibria, plus hazard, initial-flow, and stock implications for aggregate net productivity.**Expected declarations:** Parameter updates/transports; four general fixed-tightness orders and two strict positive-option-value refinements; four robust equilibrium orders; hazard/flow/stock lemmas; two M7 capstones**Adequacy grade:** GREEN **Dependencies:** M5–M6**Blocker:** None for the proved pairwise results**Branch/commit or tag; last review; next action:** issue2; M7 completion commit; reviewed 2026-07-31 by human mathematical/economic review; preserve the M7 interface and design M8’s Appendix-specific derivative layer.**M8. Appendix derivative comparative statics****Status:** COMPLETE – AMBER**Paper:** Appendix; equations (A1)–(A12), plus equation (11)**Economic target:** Derivative identities and signs along supplied differentiable equilibrium paths.**Expected declarations:** Appendix matching assumptions; CDF/expected-excess analytic lemmas; exact path structures; (11), (A1)–(A12); component and overall capstones**Adequacy grade:** Historical supplied-path layer AMBER; analytic statements remain valid **Dependencies:** M3–M4, M7 analytic interfaces**Blocker:** Paths were supplied in M8; M8b now discharges that premise locally**Branch/commit or tag; last review; next action:** issue2; M8 completion commit; reviewed 2026-08-01; preserve the historical record and use the GREEN M8b package for supplied regular equilibria.

M8b.1. Local IFT foundation and lambda path**Status:** COMPLETE – GREEN**Paper:** Appendix foundation**Economic target:** Derive fixed-tightness sigma and lambda paths through a supplied reduced equilibrium.**Expected declarations:** Appendix IFT regularity; parameterized residuals; scalar IFT wrapper; two path constructors and capstones**Adequacy grade:** GREEN relative to a supplied equilibrium under explicit primitive C1 regularity of (q)**Dependencies:** M8**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; reviewed 2026-08-01; preserve these interfaces as the foundation for completed M8b.2.**M8b.2. Discount and dispersion IFT paths****Status:** COMPLETE – GREEN**Paper:** Appendix foundation**Economic target:** Construct discount and dispersion paths through a supplied regular equilibrium.**Expected declarations:** Local C1 residuals; derived negative cutoff partials; two path constructors and capstones; full implicit Appendix capstone**Adequacy grade:** Full Appendix package GREEN relative to a supplied regular reduced equilibrium**Dependencies:** M8b.1**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; reviewed 2026-08-02; begin M9 separately with aggregate-state transition terms.**M9.1. Anticipated two-state foundations****Status:** COMPLETE – GREEN**Paper:** Section 4; coupled system and conditional equations (16)–(18)**Economic target:** Two-state primitives, primitive value equilibrium, and general max-based surplus system.**Expected declarations:** Aggregate state; two-state primitives and values; coupled/state Bellman equations; conditional regional equations; capstone**Adequacy grade:** GREEN **Dependencies:** M0–M3**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; reviewed 2026-08-02; preserve the conditional max-form representation.**M9.2. Cutoffs, ordering, and certified regions****Status:** COMPLETE – GREEN**Paper:** Section 4; cutoff-ordering discussion and regional formulas**Economic target:** Statewise cutoffs, their economic admissibility and ordering, and certified interval formulas.**Expected declarations:** Simultaneous surplus differences; monotonicity and bounds; derived cutoffs and signs; statewise interval integrals; proved ordering; regional slopes; hinge/option values; equations (16)–(24); capstones**Adequacy grade:** GREEN, conditional on a supplied TwoStateValueEquilibrium **Dependencies:** M9.1**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; M9.2A af3dbb6; M9.2B 7d65582; M9.2C completion commit; reviewed 2026-08-04; preserve the interface.**M9.3. Job creation, dynamics, and impact asymmetry****Status:** COMPLETE – GREEN**Paper:** Section 4; equation (15), equations (25)–(30)**Economic target:** Statewise job creation, unemployment dynamics, and impact asymmetry.**Expected declarations:** Equations (25)–(30); statewise hazards and flows; equation (15); stationary unemployment; measure-valued impact operator; exact interval destruction; capstones**Adequacy grade:** GREEN, conditional on a supplied value equilibrium and admissible employment distributions**Dependencies:** Reviewed M9.2, M6; Appendix matching for strict worker-hazard monotonicity**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; reviewed 2026-08-04; preserve the reviewed Section 4 package; equation (31) remains unimplemented.

M10.1. Finite-state Markov equilibrium**Status:** NOT STARTED**Paper:** Section 5 setup; equations (32)–(34)**Economic target:** Finite-state continuous-time Markov equilibrium representation.**Expected declarations:** Finite process and row-stochastic weights; next-state expectation; FiniteMarkovEquilibrium; equations (32)–(34); two-state embedding**Adequacy grade:** Target: GREEN **Dependencies:** M9.3**Blocker:** None**Branch/commit or tag; last review; next action:** No branch/tag or review; implement the representation without existence or discrete-time transition claims.**M10.2. Discrete-time employment transition****Status:** NOT STARTED**Paper:** Section 5; equation (35)**Economic target:** Measure-valued discrete-time employment transition.**Expected declarations:** Separate one-period redraw probability; transition measure; upper-support creation atom; optional density corollary**Adequacy grade:** Target: GREEN **Dependencies:** M10.1**Blocker:** Continuous-time rates must not be used as one-period probabilities**Branch/commit or tag; last review; next action:** No branch/tag or review; begin only after M10.1 review.**M10.3. Creation, destruction, and accounting****Status:** NOT STARTED**Paper:** Section 5; equations (36)–(38)**Economic target:** Creation, destruction, and employment-mass accounting.**Expected declarations:** Creation flow; impact and redraw destruction; no double counting; employment-mass identity**Adequacy grade:** Target: GREEN **Dependencies:** M10.2**Blocker:** Requires reviewed transition operator**Branch/commit or tag; last review; next action:** No branch/tag or review; begin only after M10.2 review.**NUM. Numerical simulation and calibration****Status:** DEFERRED / OUTSIDE CORE LEAN**Paper:** Section 5; equations (39)–(42), Tables I–II**Economic target:** External Python or Julia numerical replication; Lean only for exact accounting or certified residuals where useful.**Expected declarations:** External solver and optional Lean residual/accounting certificates**Adequacy grade:** – **Dependencies:** Analytical interfaces as needed**Blocker:** Numerical/empirical work is outside core Lean**Branch/commit or tag; last review; next action:** No branch/tag or review; define a reproducible calibration and residual protocol externally.

Adequacy notes. M6: docs/static_steady_state_adequacy.md. Inherited M5 existence foundation: docs/static_existence_foundation.md. M7 scope: docs/static_comparative_statics_scope.md. M8 scope: docs/appendix_differentiability_scope.md.

M0 acceptance checklist

- ☒ New module root is isolated from the legacy architecture.
- ☒ Primitive parameter structure exists.
- ☒ Economic assumptions are layered.
- ☒ Shock law is represented by a measure.
- ☒ CDF is derived from the measure.
- ☒ Matching rates are defined.
- ☒ Surplus is defined from J, W, U , not stored independently.
- ☒ Equations (1)–(7) are represented faithfully.
- ☒ Equations (5) and (6) use one shared continuation term.
- ☒ No cutoff or reduced-equilibrium conclusion is assumed.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, or custom axioms exist.
- ☒ Tracker TeX compiles and PDF has been visually inspected.
- ☒ Human mathematical/economic review is complete.

M1 acceptance checklist

- ☒ Generic continuation decomposition is proved under a probability measure.
- ☒ Firm surplus share is derived from equations (3) and (4).
- ☒ The surplus flow equation is derived from equations (5)–(7).
- ☒ Equation (8) is proved for the actual equilibrium surplus.
- ☒ The main theorem uses only probability normalization and existing integrability.
- ☒ The **ShockAssumptions** wrapper installs the probability instance locally.
- ☒ No M0 structure was strengthened or modified.
- ☒ No cutoff, affine surplus, or reduced equilibrium was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, or custom axioms exist.
- ☒ Proof ledger TeX/PDF compiles and passes visual inspection.
- ☒ Milestone tracker TeX/PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M2 acceptance checklist

- ☒ Equation (8) is subtracted at two states.
- ☒ The scaled two-point surplus identity is proved.
- ☒ The divided affine-surplus identity is proved.
- ☒ Surplus is proved strictly increasing.
- ☒ The cutoff is defined from actual equilibrium surplus.
- ☒ The cutoff is proved to be a surplus zero.
- ☒ The zero is proved unique.
- ☒ $V = 0$ is derived from free entry and $r > 0$.
- ☒ $q(\theta)J(\varepsilon_u) = c$ follows from (1)–(2).
- ☒ $J(\varepsilon_u) > 0$ is proved.
- ☒ $S(\varepsilon_u) > 0$ is proved.
- ☒ $\varepsilon_d < \varepsilon_u$ is proved.
- ☒ Equation (12) is proved.
- ☒ Surplus signs are characterized relative to ε_d .
- ☒ Firm-value signs are characterized relative to ε_d .
- ☒ Active surplus is rewritten as a scaled positive part.
- ☒ No M0 or M1 mathematical declaration was changed.
- ☒ No M3 or later theorem was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, or custom axioms exist.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M3 acceptance checklist

- ☒ Expected excess is the integral of the positive part.
- ☒ Tail option value is defined from the induced CDF.
- ☒ Expected-excess integrability is proved.
- ☒ Strict upper-tail probability equals one minus the CDF.
- ☒ The strict-tail layer-cake identity is proved.
- ☒ Active-surplus expectation is rewritten using expected excess.
- ☒ Measure-form equation (9) is derived.
- ☒ Paper-facing equation (9) is derived.
- ☒ The free-entry search-gain identity is derived.
- ☒ Measure-form equation (10) is derived.
- ☒ Paper-facing equation (10) is derived.
- ☒ Job-destruction residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-destruction predicate.
- ☒ The product-form job-creation identity is derived.
- ☒ Paper equation (13) is derived.
- ☒ Job-creation residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-creation predicate.
- ☒ No **ReducedEquilibrium** structure is introduced.
- ☒ No M0–M2 mathematical declaration is changed.
- ☒ No M4 or later theorem is introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, custom axiom, or opaque placeholder exists.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M4 acceptance checklist

- ☒ Robust product-form job-creation predicate is defined.
- ☒ Reduced pair stores positive theta and an admissible cutoff.
- ☒ Measure-form JD and product-form JC are the only stored equations.
- ☒ Generic positive-part integrability and tail identity are proved.
- ☒ Reduced paper equations (10) and (13) are derived.
- ☒ Every value equilibrium maps to a reduced equilibrium.
- ☒ Candidate surplus, J, U, W, V, and explicit Nash wage are defined.
- ☒ Active surplus is integrable and candidate equation (8) is proved.
- ☒ Reconstructed equations (1)–(7) are proved.
- ☒ A reduced pair reconstructs a value equilibrium and its cutoff.
- ☒ Reconstruct–reduce is exact; reverse economic components agree.
- ☒ Conditional nonempty equivalence is proved.
- ☒ No unconditional existence, uniqueness, M0–M3 change, or M5 theorem.
- ☒ Targeted and full builds pass; audits and PDFs pass.
- ☒ Human mathematical/economic review is complete.

M5 acceptance checklist

- ☒ Expected excess is antitone, one-Lipschitz, and continuous.
- ☒ JD net value and JD-implied tightness are defined.
- ☒ JD satisfaction is equivalent to $\theta = \theta_{JD}(d)$.
- ☒ The JD curve is strictly increasing.
- ☒ The scalar JC residual is defined and strictly decreasing on its admissible domain.
- ☒ Figure 1 JD-upward and JC-downward slope theorems are proved.
- ☒ At most one `ReducedEquilibrium` is proved.
- ☒ `StaticExistenceAssumptions` contains no root or equilibrium witness.
- ☒ Bracket continuity and upper residual $-c$ are proved.
- ☒ An interior root and a reduced witness are constructed by the IVT.
- ☒ Conditional unique reduced existence is proved.
- ☒ Value nonemptiness, theta/cutoff uniqueness, and economic uniqueness follow through M4.
- ☒ Existence is labeled conditional on the additional bracket assumptions.
- ☒ No M0–M4 theorem statement changed and no M6 theorem was introduced.
- ☒ Targeted/full builds, audits, and both PDFs pass.
- ☒ Human mathematical/economic review is complete.

M6 acceptance checklist

- ☒ Strict destruction mass matches the M2 rule and equals the CDF under no atoms.
- ☒ Separation/job-finding hazards, creation/destruction flows, and the residual are defined.
- ☒ Total transition hazard is positive.
- ☒ Closed-form unemployment is in $[0, 1]$, balances flows, and is unique.
- ☒ Paper equation (14) is proved.
- ☒ **SteadyStateEquilibrium** is defined; employment and vacancies are derived.
- ☒ Creation is $q(\theta)v$; destruction is $\lambda F(d)(1 - u)$; flows are equal.
- ☒ Positive separation implies positive unemployment and vacancies.
- ☒ $v/u = \theta$ is proved only under positivity.
- ☒ Every reduced equilibrium completes to a full state; exact round trips and an equivalence are proved.
- ☒ At most one full state is proved without existence assumptions.
- ☒ Conditional nonemptiness and unique existence are transported from M5.
- ☒ No Beveridge-curve slope, M0–M5 statement change, or M7+ theorem was introduced.
- ☒ Targeted/full builds, no-sorry/axiom audits, and prohibited-pattern searches pass.
- ☒ Proof-ledger and tracker PDFs compile and pass visual inspection.
- ☒ Human mathematical/economic review is complete.

M7 acceptance checklist

- ☒ Parameter-update functions are defined and assumption transport is proved.
- ☒ Fixed-tightness cutoff results for p, b, λ, r are proved.
- ☒ Full-equilibrium p and b orders are proved.
- ☒ Full-equilibrium λ cutoff and r tightness orders are proved.
- ☒ Separation-hazard and job-finding-rate orders are proved.
- ☒ Initial creation/destruction flow results are proved.
- ☒ Steady unemployment and employment orders are proved.
- ☒ The vacancy response is explicitly left ambiguous.
- ☒ No core theorem uses **StaticExistenceAssumptions**.
- ☒ M8 derivative results remain unimplemented.
- ☒ No M0–M6 mathematical theorem changed.
- ☒ Targeted and full builds pass.
- ☒ No **sorry**, **admit**, custom axiom, or opaque placeholder exists.
- ☒ Proof-ledger and tracker PDFs pass visual inspection.
- ☒ Human mathematical/economic review is complete.

M8 acceptance checklist

- ☒ Appendix assumptions are isolated from the core.
- ☒ Appendix matching elasticity and $0 < \eta < 1$ are explicit.
- ☒ CDF continuity and the expected-excess derivative are derived from the shock law.
- ☒ Weak/strict tail-gap bounds and moment identities use explicit shock assumptions.
- ☒ Path structures contain exact differentiability and closure, but no sign conclusion.
- ☒ Equation (11), its sign characterization, and (A1)–(A12) are proved.
- ☒ Lambda cutoff and tightness derivatives are negative.
- ☒ Discount tightness is negative and the cutoff has the exact ambiguous (A8) condition.
- ☒ Dispersion derivatives are positive under $b \leq p$.
- ☒ The lambda capstone explicitly takes $0 < \lambda$.
- ☒ No result takes **StaticExistenceAssumptions** or selects an M5 witness.
- ☒ M8's AMBER source is supplied-path regularity, not M5 inheritance.
- ☒ The six-step M8b route is documented; no M0–M7 theorem changed.
- ☒ The M8 review surface was isolated from later M8b and Section 4 work.
- ☒ All ten M8 sources pass direct elaboration; aggregate and full builds pass.
- ☒ No sorry, admit, custom axiom, or opaque placeholder exists.
- ☒ Transitive axiom checks contain no custom project axiom.
- ☒ Both PDFs pass visual inspection and the review archive excludes build artifacts.
- ☒ Human mathematical/economic review is complete (2026-08-01).

M8b.1 acceptance checklist

- ☒ Appendix IFT assumptions contain only local C1 regularity of (q).
- ☒ The installed Mathlib IFT API is recorded.
- ☒ Expected-excess local C1 regularity is derived.
- ☒ Both base roots and cutoff nondegeneracy results are proved.
- ☒ The generic wrapper exposes local closure and graph uniqueness.
- ☒ Fixed-sigma and lambda paths and their M8 capstones are constructed.
- ☒ No M5-selected equilibrium or static-existence assumption is used.
- ☒ The reviewed M8b.1 layer contains no M8b.2 or Section 4 theorem.
- ☒ Builds and source audits pass; PDFs are visually inspected.
- ☒ Human mathematical/economic review is complete (2026-08-01).

M8b.2 acceptance checklist

- ☒ Discount and dispersion base roots are derived from the supplied equilibrium.
- ☒ Both cutoff partials are proved strictly negative and nonzero.
- ☒ Joint local C1 regularity is proved for both JD graphs and residuals.
- ☒ Both equilibrium paths are constructed by the scalar IFT.
- ☒ Local positivity, JD, product-form JC, and base recovery are proved.
- ☒ Discount preserves the exact A8 iff; dispersion signs use normalization and $b \leq p$.
- ☒ The full capstone takes no path, M5 selection, sign, or nondegeneracy assumption.
- ☒ No Section 4 declaration is implemented.
- ☒ Direct, aggregate, audit, and full builds pass.
- ☒ Source audits and PDF visual inspection pass.
- ☒ Human mathematical/economic review is complete (2026-08-02).

M9.1 acceptance checklist

- ☒ **AggregateState** is defined.
- ☒ State switching is involutive.
- ☒ Two-state productivity primitives are defined.
- ☒ High productivity and positive transition rate are primitive assumptions.
- ☒ No cutoff ordering is assumed.
- ☒ Two-state value candidates are defined.
- ☒ Statewise surplus is derived from (J,W,U).
- ☒ Statewise active-surplus integrability is required.
- ☒ Vacancy and unemployment equations include aggregate capital gains.
- ☒ Firm and worker aggregate continuation terms are economically correct.
- ☒ Firm share and zero vacancy values are derived.
- ☒ Raw and general coupled max-form surplus equations are derived.
- ☒ Recession and boom max-form equations are derived.
- ☒ Conditional equations (16), (17), and (18) are derived.
- ☒ No cutoff or cutoff ordering is introduced.
- ☒ No equation (15) or equations (19)–(30) are introduced.
- ☒ No M0–M8b.2 theorem changes.
- ☒ Targeted, aggregate, audit, and full builds pass.
- ☒ No placeholder or custom axiom exists.
- ☒ Both PDFs pass visual inspection.
- ☒ Human mathematical/economic review is complete (2026-08-02).

M9.2 acceptance checklist

- ☒ Coupled two-point surplus identities and simultaneous strict monotonicity are derived.
- ☒ Lower and upper increment bounds and Lipschitz continuity are proved.
- ☒ Positive upper surplus and negative witnesses are proved statewise.
- ☒ Unique admissible cutoffs and complete reservation signs are derived.
- ☒ Active-surplus expectations are rewritten over statewise survival intervals.
- ☒ No cross-state cutoff order is assumed.
- ☒ Conditional interval geometry under either order is checked.
- ☒ The initial ordering obstruction is preserved as historical context.
- ☒ Human review identified the stronger maximum-principle route.
- ☒ The strict order $d_B < d_R$ is derived from reviewed assumptions.
- ☒ Exact ordered equation (16) is proved.
- ☒ Exact ordered equation (17) is proved.
- ☒ Exact ordered equation (18) is proved.
- ☒ Regional difference formulas and equation (21) are proved.
- ☒ Recession option-value formula is proved.
- ☒ Boom piecewise option-value formula is proved.
- ☒ Equations (19), (20), (22), (23), and (24) are proved.
- ☒ No M9.3 theorem is implemented.
- ☒ Direct, aggregate, audit, and full builds pass.
- ☒ Source and axiom audits pass.
- ☒ Documentation PDFs compile and pass visual inspection.
- ☒ Human mathematical/economic review of M9.2C/full M9.2 is complete (2026-08-04).
- ☒ M9.2A human mathematical/economic review is complete (2026-08-02).
- ☒ M9.2B human mathematical/economic review is complete (2026-08-04).

M9.3 acceptance checklist

- ☒ Equation (25).
- ☒ Equation (26).
- ☒ Equation (27).
- ☒ Equation (28).
- ☒ Equation (29).
- ☒ Boom denominator positivity.
- ☒ Equation (30).
- ☒ Anticipation wedge.
- ☒ Vacancy and worker meeting rates are distinguished.
- ☒ Statewise destruction hazards are defined.
- ☒ Worker-meeting hazard is increasing on positive tightness.
- ☒ Boom worker-meeting hazard exceeds recession.
- ☒ Boom idiosyncratic destruction hazard is weakly lower.
- ☒ Equation (15) vector field is stated.
- ☒ Stationary unemployment and drift signs are proved.
- ☒ Cross-state drift order is proved.
- ☒ Employment is represented by a measure.
- ☒ Upturn impact employment is unchanged.
- ☒ Downturn destroys exactly $[d_B, d_R]$.
- ☒ Strict downturn impact requires positive interval mass.
- ☒ The no-impact-creation timing convention is explicit.
- ☒ No density assumption is used.
- ☒ No M10 theorem is implemented.
- ☒ Direct and aggregate builds pass.
- ☒ No placeholder or custom axiom exists.
- ☒ Both PDFs are visually inspected.
- ☒ Human mathematical/economic review is complete (2026-08-04).

M5b foundational follow-up

Status: NOT STARTED. The adequacy target is to upgrade static existence from amber to green under either a strengthened general matching-boundary theorem or a fully certified Cobb–Douglas specialization. M5b is a foundational refinement, not M6, and is not part of the current M5 Lean proof.

The proposed work adds a right-hand Inada condition for q , separately requires $b < p + \sigma \varepsilon_u$, proves that the JD curve reaches zero and then positive tightness, derives `lower_crossing`, and constructs `StaticExistenceAssumptions`. The Inada condition on q alone is insufficient because it says nothing about whether JD-implied tightness ever becomes positive. The complete design is in `docs/static_existence_foundation.md`. M6 does not implement this refinement.

Changelog

2026-08-04 – M9.3 human review completed. Graded equations (25)–(30), hazards and equation (15), stationary unemployment algebra, cross-state flows, the impact operator, and the full asymmetry theorem COMPLETE – GREEN. Confirmed equation (27)’s reuse of M9.2C, the labor-flow/ODE boundary, the three rate concepts, exact $[d_B, d_R]$ downturn destruction and its strict-mass qualification, and zero instantaneous upturn creation. Full M9.1–M9.3 is COMPLETE – GREEN conditional on a supplied two-state equilibrium. Equation (31) remains unimplemented; M10.1 is next.

2026-08-04 – M9.3 cyclical dynamics implemented. Derived equations (25)–(30), denominator positivity, and the anticipation wedge. Defined statewise hazards, equation (15)’s vector field, stationary unemployment, and the cross-state drift comparison without claiming ODE existence. Added the density-free employment-measure impact operator: upturn employment is unchanged, downturn destruction is exactly $[d_B, d_R]$, strict loss requires positive interval mass, and impact creation is zero by timing convention. M9.3 is READY FOR REVIEW, target COMPLETE – GREEN; M10 remains NOT STARTED.

2026-08-04 – M9.2C and full M9.2 human review completed. Graded equations (16)–(24), the regional difference and hinge/option-value layers, and the M9.2C capstone COMPLETE – GREEN. Graded full M9.2 COMPLETE – GREEN, conditional on a supplied `TwoStateValueEquilibrium`. Confirmed that no order, affinity, derivative, interval equation, or option-value formula is stored in the equilibrium and that no `StaticExistenceAssumptions`

or primitive-selected equilibrium is used. M9.3 is next and remains NOT STARTED.

2026-08-04 – M9.2C ordered regional equations implemented. Derived exact ordered equations (16)–(18), regional difference formulas, equations (21) and (23), both hinge and option-value identities, and equations (19), (20), (22), and (24). Added both M9.2 capstones and aggregate/audit coverage. M9.2C is READY FOR REVIEW, target COMPLETE – GREEN; M9.3 remains NOT STARTED.

2026-08-04 – M9.2B human review completed. Graded the contrary-order comparison, reverse-order contradiction, strict cutoff order, and strict upper-surplus/tightness implications COMPLETE – GREEN. Confirmed $d_B < d_R$ is derived from a supplied equilibrium without an order, affinity, derivative, search-gain, or closure assumption. Confirmed the maximum-principle chain $S_B \leq S_R$, $\theta_B \leq \theta_R$, $D_u \leq I_B - I_R$, and $(r + 2\mu)D_u \geq p_H - p > 0$ while $D_u \leq 0$. M9.2C is next.

2026-08-02 – M9.2B maximum-principle cutoff ordering implemented. Proved regional cross-state gap constancy under the contrary weak cutoff order, the global reverse-order surplus comparison, and the induced weak tightness order from free entry. Proved the pointwise and integrated active-gap bounds and exposed the upper-support gap equation. Derived `boom_cutoff_lt_recession_cutoff` without an added order, surplus, tightness, search-gain, or closure assumption, plus strict boom upper-surplus and tightness orders. M9.2B is READY FOR REVIEW; M9.2C and M9.3 remain NOT STARTED.

2026-08-02 – M9.2A human review completed. Simultaneous monotonicity, global bounds and continuity, unique cutoffs, reservation signs, and statewise interval integrals are COMPLETE – GREEN. No lower shock support or cross-state order is used, and the results are correctly conditional on a supplied equilibrium. Human review identified the stronger maximum-principle route for M9.2B, now NOT STARTED with a reviewed proof route available.

2026-08-02 – M9.2A implemented; initial ordering route stalled. Derived simultaneous monotonicity, increment bounds, continuity, positive upper surplus, negative witnesses, unique admissible cutoffs, reservation signs, and statewise active-surplus interval integrals. Conditional interval geometry was checked under either order. The endogenous tightness-weighted search-gain difference remained unsigned relative to productivity and continuation terms. No ordering assumption was added. This initial classification was superseded by the review-discovered maximum-principle route recorded above.

2026-08-02 – M9.1 human review completed. Review confirmed a conditional representation theorem for every supplied two-state value equilibrium; existence is not asserted and does not make the result AMBER. Aggregate capital gains are present in all four primitive value equations, and the coupled surplus equation is derived. Equations (16)–(18) are max-form sign reductions rather than final cutoff-indexed interval formulas. No cutoff or ordering is assumed. M9.1 is COMPLETE – GREEN; M9.2 is next and NOT STARTED.

2026-08-02 – M9.1 anticipated two-state foundations implemented. Added six cyclical modules for the recession/boom process, primitive value system, shared continuation terms, general coupled max-based surplus equation, conditional equations (16)–(18), and capstone. Probability remains explicit through `ShockAssumptions`; all results are conditional on a supplied `TwoStateValueEquilibrium`. No cutoff, ordering, existence, job-creation, unemployment-dynamics, or impact-asymmetry theorem was introduced. M9.1 is READY FOR REVIEW, target GREEN; M9.2 and M9.3 remain NOT STARTED.

2026-08-02 – M8b.2 human review completed. The discount residual/path, dispersion residual/path, and full IFT-derived Appendix capstone are COMPLETE – GREEN. Roots and strictly negative cutoff partials are derived, all four paths are constructed through Mathlib's IFT, and no path, sign, M5 selection, or static-existence assumption is used. The full Appendix package is GREEN relative to a supplied regular reduced equilibrium and explicit Appendix regularity/sign conditions. Primitive base selection would separately inherit M5 AMBER. M9 is next and remains NOT STARTED.

2026-08-01 – M8b.2 discount and dispersion paths implemented. Derived both base roots and strictly negative cutoff partials from a supplied reduced equilibrium, proved joint local C1 regularity, constructed both local equilibrium paths by the scalar IFT, and connected them to the reviewed M8 discount and dispersion results. The full implicit Appendix capstone has no M5, supplied-path, sign, or nondegeneracy premise. M8b.2 is READY FOR REVIEW, target GREEN; overall M8 remains COMPLETE – AMBER pending review.

2026-08-01 – M8b.1 human review completed. The generic wrapper, fixed-sigma path, lambda path, and combined result are COMPLETE – GREEN relative to a supplied equilibrium under explicit Appendix IFT regularity. Base roots and nondegeneracy are proved; no sign or M5 selection is assumed. The C1 condition on (q) is a technical strengthening, not a paper conclusion, and no global uniqueness is claimed. Overall M8 stays COMPLETE – AMBER pending M8b.2 review.

2026-08-01 – M8b.1 local IFT foundation implemented. Added primitive matching C1 regularity, parameterized residual calculus, the generic scalar Mathlib IFT wrapper, and derived fixed-tightness sigma and lambda equilibrium paths. Base roots and nonzero cutoff derivatives are proved; none is assumed. No M5 existence result is used. M8b.1 is READY FOR REVIEW, target GREEN; M8 remains COMPLETE – AMBER until M8b.2.

2026-08-01 – M8 human review completed. The Appendix identities and signs were found faithful and non-circular. Analytic infrastructure is COMPLETE – GREEN; supplied-path results and overall M8 are COMPLETE – AMBER. The AMBER source is supplied differentiable local paths, not M5. The lambda capstone uses $0 < \lambda$, and M8b remains the route to GREEN for the path layer.

2026-08-01 – M8 Appendix derivative comparative statics integrated. Added isolated Appendix assumptions, shock-law analytic lemmas, exact supplied paths, scalar algebra, equations (11) and (A1)–(A12), sign theorems, and capstones. Lambda lowers cutoff and tightness; discount lowers tightness and has the exact (A8) cutoff condition; dispersion raises both derivatives under $b \leq p$. No M8 theorem uses `StaticExistenceAssumptions` or an M5 witness. The missing IFT path construction is the sole AMBER gap and is deferred to M8b. M8 is READY FOR REVIEW, overall COMPLETE – AMBER; human review remains pending.

2026-07-31 – M7 human review completed. The order proofs were found mathematically and economically sound; M7 is COMPLETE – GREEN. No M5 existence assumption is used. Unnecessary imports of the M5 existence and M6 transported-existence layers were removed. Strict fixed-tightness refinements were added under explicit positive option value, with positive λ additionally required for the strict discount-rate result. Proof-ledger symbol and rendering defects were corrected. The λ -tightness, r -cutoff, σ , and derivative results remain M8, which is NOT STARTED.

2026-07-31 – M7 static order comparative statics implemented. Added immutable parameter updates and transports, four fixed-tightness cutoff orders, robust supplied-equilibrium orders for p, b, λ, r , and weak hazard, flow, unemployment, and employment implications for aggregate net productivity. Vacancies remain ambiguous. No M7 core result uses `StaticExistenceAssumptions`; existence is not asserted. The λ -tightness, r -cutoff, σ , and Appendix derivative results remain M8. Added `docs/static_comparative_statics_scope.md`. M7 is **READY FOR REVIEW**, target **COMPLETE – GREEN**.

2026-07-31 – M6 human review completed. Stock completion, equation (14), accounting, exact representation, and full-state uniqueness were judged mathematically and economically faithful and are **GREEN**. Zero separation correctly yields $u = v = 0$, with the literal ratio guarded by positive separation. M6 adds no existence closure premise; existence inherits M5's `lower_crossing` and remains **AMBER**. Added split green/amber capstones and the detailed record `docs/static_steady_state_adequacy.md`. M6 is **COMPLETE – AMBER**; M7 is **NOT STARTED**.

2026-07-31 – M6 unemployment flow balance implemented. Added the strict destruction mass and atomless CDF bridge, hazards, flows, the unique balanced unemployment formula, and equation (14). Added the full static state with derived employment/vacancies, guarded $v/u = \theta$ by positive separation, and proved exact reduced/full equivalence, uniqueness, and conditional existence inherited from M5. Stock completion is recommended **GREEN**; existence and overall M6 were submitted as **AMBER** pending the completed review above.

2026-07-31 – M5 static-existence foundation documented. Recorded the exact current continuity and lower-crossing assumptions and the unimplemented M5b route combining right-hand Inada behavior with separate upper-job profitability. The note makes explicit that Inada alone is insufficient. M5 remains **COMPLETE – AMBER**; M6 remains **NOT STARTED**. The durable design note is `docs/static_existence_foundation.md`.

2026-07-31 – M5 review completed. The uniqueness and IVT proofs passed review. Figure 1 establishes uniqueness, not existence. The non-circular `lower_crossing` field assumes a positive residual rather than a root, but supplies the missing endpoint condition. Existence-only signatures now omit matching assumptions. Uniqueness is **COMPLETE – GREEN**; existence and overall M5 are **COMPLETE – AMBER**. A future task is to derive the lower crossing from primitive conditions on q , or certify it for Cobb–Douglas; this does not block M6.

2026-07-31 – M5 static existence and uniqueness implemented. Expected excess is one-Lipschitz; JD slopes upward and JC downward, proving at most one reduced equilibrium. The separate continuity/lower-bracket bundle and the derived upper residual $-c < 0$ yield an interior crossing by the IVT. M4 transports conditional existence and economic uniqueness to value equilibria. The implementation was submitted for review; equation (14) remains M6.

2026-07-31 – M4 review completed. Review found the generic excess and CDF-tail analysis valid, and equations (10) and (13) sufficient for reconstruction under the stated admissibility assumptions. The Nash wage is economically correct; equations (1)–(7) are reconstructed; and both economic round trips are certified. The import diagram and assumption-field account were corrected. M4 proves no existence or uniqueness theorem and is **COMPLETE – GREEN**.

2026-07-30 – M4 reconstruction and equivalence implemented. Created the robust reduced pair, generic analytic layer, forward bridge, explicit reverse reconstruction, and equivalence module. Equations (1)–(7) are reconstructed with the Nash wage; the reduced round trip is exact and the value round trip is componentwise exact. Nonemptiness equivalence consumes a witness on either side and proves no existence or uniqueness. M4 is **READY FOR REVIEW**.

2026-07-30 – M3 review completed. The layer-cake/CDF proof was reviewed and found valid without atomlessness. Equations (9), (10), and (13) have the correct signs and dependencies, and job creation is independent of the job-destruction equation. The capstone was moved to the neutral combining module `SteadyState/StaticConditions.lean`. Residual predicates require separate admissibility fields in M4, which must also prove a generic expected-excess/tail theorem for reverse reconstruction. M3 was graded **COMPLETE – GREEN**.

2026-07-30 – M3 continuation and static conditions implemented. Created `SteadyState/Continuation.lean`, `SteadyState/JobDestruction.lean`, and `SteadyState/JobCreation.lean`, with the capstone in the neutral `SteadyState/StaticConditions.lean`. Expected excess follows from the M2 active-surplus formula; Mathlib's strict-tail layer-cake theorem gives the CDF representation without atomlessness. Equations (9), (10), and (13), the two residual predicates, and the forward M3 capstone are proved for an existing `ValueEquilibrium`. No reduced-equilibrium structure or M4 theorem was introduced.

2026-07-29 – M2 review completed. Review confirmed that equation (8) was correctly subtracted at two states and that actual equilibrium surplus is affine and strictly increasing. The derived `reservationCutoff` is its uniquely identified zero; equations (1)–(2) establish $\varepsilon_d < \varepsilon_u$; and equation (12) and the reservation rule are proved. M2 remains conditional on an existing `ValueEquilibrium` and was graded **COMPLETE – GREEN**.

2026-07-29 – M2 affine surplus and cutoff implemented. Created `SteadyState/AffineSurplus.lean` and `SteadyState/Cutoff.lean`. Subtracting equation (8) gives affine surplus and strict monotonicity. The derived cutoff is the unique surplus zero; equations (1)–(2), positive vacancy cost and meeting, and $\beta < 1$ separately place it below ε_u . Added equation (12), reservation sign rules, the active-surplus bridge, and the M2 capstone. M2 is **READY FOR REVIEW**.

Review-archive convention. Beginning with M3, project-local review archives are named `review_m3.zip`, `review_m4.zip`, and so on. They remain untracked and are never committed.

2026-07-29 – M1 review completed. Human mathematical and economic review found equation (8) mathematically faithful to the paper and its proof non-circular. It is derived from equations (3)–(7), uses the actual equilibrium surplus, and assumes no cutoff, affinity result, or reduced equilibrium. It is a conditional representation theorem for an existing `ValueEquilibrium`; M1 does not prove equilibrium existence. M1 was graded **COMPLETE – GREEN**.

2026-07-29 – M1 equation (8) implemented. Created `MP1994V2/SteadyState/Surplus.lean` with the continuation decomposition, firm share, surplus flow equation, minimal probability theorem, and `ShockAssumptions` wrapper. Updated `All`, `Audit`, `README`, trackers, and added the proof ledger. M1 is **READY FOR REVIEW**.

2026-07-29 – M0 review completed. Review found equations (1)–(7) economically faithful and no future conclusion embedded; corrected substantive modules $\rightarrow$ `All` $\rightarrow$ `Audit`; required explicit `ShockAssumptions` `P` and local `D.isProbability` in M1; graded **COMPLETE – GREEN**.

2026-07-29 – v2 foundation created. Added `Definitions`, `Assumptions`, `Probability`, `Matching`, `Equilibrium/Value`, `All`, `Audit`, `README`, and synchronized trackers: primitives, four assumption bundles, CDF/support/matching notation, shared surplus/continuation, `ValueCandidate`, and `ValueEquilibrium`. No legacy v1 Lean or architecture file changed; the library-root import makes `lake build certify MP1994V2.All`.